

Scripting Basics: Python

Practical task

Task 1

1. Create a script that accepts the file name and puts its extension to output. If there is no extension - an exception should be raised.
2. Create a script that reads the access log from a file. The name of the file is provided as an argument. An output of the script should provide the total number of different User Agents and then provide statistics with the number of requests from each of them. [Here is a link to an example access.log file.](#)
3. Develop a python script that:
 - Creates an rc tag for the latest commit in specific Github repository for main branch
 - Creates a release tag when executed with a `--create-release` flag
 - The rc tag should follow the format `vX.Y.Z`(e.g: `rc/1.2.3`). Y increments with each new rc tag created
 - If run with `--create-release` flag: Create rc tag and release tag in format `X.Y.Z`
4. Write a script that gets system information like distro info, memory(total, used, free), CPU info (model, core numbers, speed), current user, system load average, and IP address. Use arguments for specifying resources. (For example, `-d` for distro `-m` for memory, `-c` for CPU, `-u` for user info, `-l` for load average, `-i` for IP address).

Task2

Create a script that uses the Survey Monkey (<https://www.surveymonkey.com>) service, and creates a survey.

PREREQUISITES:

Sign up at <https://www.surveymonkey.com>

Create a draft application at <https://developer.surveymonkey.com>

No need to deploy your application. It's just for testing. Do not forget to set permissions for your application.

After creating a draft application you will obtain an

`ACCESS_TOKEN` which is needed to do API requests from your script.

REQUIREMENTS:

The script should accept a JSON file with questions for the survey and a text file with a list of email addresses.

The structure of a JSON file with questions:

```
{
  "Survey_Name": {
    "Page_Name": {
      "Question1_Name": {
        "Description" : "Description of question",
        "Answers" : [
          "Answer1",
          "Answer2",
          "Answer3"
        ]
      },
      "Question2_Name": {
        "Description" : "Description of question",
        "Answers" : [
          "Answer1",
          "Answer2",
          "Answer3"
        ]
      }
    }
  }
}
```

iii. There should be at least 3 questions and 2 recipients.

Task1:

1. Create a script that accepts the file name and puts its extension to output. If there is no extension - an exception should be raised.

Output:

```
rpavani@C19536 Scripting Basics: Python-practical task % cd Task-1
rpavani@C19536 Task-1 % python3 file_extension.py test.py
Extension: py
```

2. Create a script that reads the access log from a file. The name of the file is provided as an argument. An output of the script should provide the total number of different User Agents and then provide statistics with the number of requests from each of them.

Output:

```
rpavani@C19536 Downloads % ls apache-tomcat-9.0.117/logs
catalina.2026-04-17.log      localhost.2026-04-20.log
catalina.2026-04-20.log      localhost_access_log.2026-04-17.txt
catalina.out                localhost_access_log.2026-04-20.txt
host-manager.2026-04-17.log  manager.2026-04-17.log
host-manager.2026-04-20.log  manager.2026-04-20.log
Localhost.2026-04-17.log

rpavani@C19536 Downloads % tail apache-tomcat-9.0.117/logs/localhost_access_log.2026-04-20.txt
0:0:0:0:0:0:1 - - [20/Apr/2026:10:11:17 +0530] "GET /asf-logo-wide.svg HTTP/1.1" 200 7089
0:0:0:0:0:0:1 - - [20/Apr/2026:10:11:17 +0530] "GET /bg-nav.png HTTP/1.1" 200 1401
0:0:0:0:0:0:1 - - [20/Apr/2026:10:11:17 +0530] "GET /bg-middle.png HTTP/1.1" 200 1918
0:0:0:0:0:0:1 - - [20/Apr/2026:10:11:17 +0530] "GET /bg-button.png HTTP/1.1" 200 713
0:0:0:0:0:0:1 - - [20/Apr/2026:10:11:17 +0530] "GET /bg-upper.png HTTP/1.1" 200 3103
0:0:0:0:0:0:1 - - [20/Apr/2026:10:12:31 +0530] "GET /jenkins HTTP/1.1" 404 759
0:0:0:0:0:0:1 - - [20/Apr/2026:10:12:31 +0530] "GET /jenkins HTTP/1.1" 404 759
0:0:0:0:0:0:1 - - [20/Apr/2026:10:15:05 +0530] "GET / HTTP/1.1" 200 11232
0:0:0:0:0:0:1 - - [20/Apr/2026:10:16:33 +0530] "GET /jenkins HTTP/1.1" 404 759
0:0:0:0:0:0:1 - - [20/Apr/2026:10:16:34 +0530] "GET /jenkins HTTP/1.1" 404 759
rpavani@C19536 Downloads % cd ~/Downloads/Assignments/"Scripting Basics: Python-practical task"/Task-1
rpavani@C19536 Task-1 % python3 log_parser.py
~/Downloads/apache-tomcat-9.0.117/logs/localhost_access_log.2026-04-20.txt
Total unique User Agents: 1

Unknown -> 25
rpavani@C19536 Task-1 % nano ~/Downloads/apache-tomcat-9.0.117/conf/server.xml
rpavani@C19536 Task-1 % cd ~/Downloads/apache-tomcat-9.0.117/bin
rpavani@C19536 bin % ./shutdown.sh
Using CATALINA_BASE:   /Users/rpavani/Downloads/apache-tomcat-9.0.117
Using CATALINA_HOME:   /Users/rpavani/Downloads/apache-tomcat-9.0.117
Using CATALINA_TMPDIR: /Users/rpavani/Downloads/apache-tomcat-9.0.117/temp
Using JRE_HOME:        /opt/homebrew/Cellar/openjdk@17/17.0.18/libexec/openjdk.jdk/Contents/Home
Using CLASSPATH:
/Users/rpavani/Downloads/apache-tomcat-9.0.117/bin/bootstrap.jar:/Users/rpavani/Downloads/apache-tomcat-9.0.117/bin/tomcat-juli.jar
Using CATALINA_OPTS:
NOTE: Picked up JDK_JAVA_OPTIONS:  --add-opens=java.base/java.lang=ALL-UNNAMED
--add-opens=java.base/java.lang.invoke=ALL-UNNAMED --add-opens=java.base/java.lang.reflect=ALL-UNNAMED
--add-opens=java.base/java.io=ALL-UNNAMED --add-opens=java.base/java.util=ALL-UNNAMED
--add-opens=java.base/java.util.concurrent=ALL-UNNAMED --add-opens=java.rmi/sun.rmi.transport=ALL-UNNAMED
Apr 22, 2026 2:11:04 PM org.apache.catalina.startup.Catalina stopServer
SEVERE: Could not contact [localhost:8005] (base port [8005] and offset [0]). Tomcat may not be running.
Apr 22, 2026 2:11:04 PM org.apache.catalina.startup.Catalina stopServer
SEVERE: Error stopping Catalina
java.net.ConnectException: Connection refused
    at java.base/sun.nio.ch.Net.connect0(Native Method)
    at java.base/sun.nio.ch.Net.connect(Net.java:591)
    at java.base/sun.nio.ch.Net.connect(Net.java:580)
    at java.base/sun.nio.ch.NioSocketImpl.connect(NioSocketImpl.java:593)
    at java.base/java.net.SocksSocketImpl.connect(SocksSocketImpl.java:327)
```

```
at java.base/java.net.Socket.connect(Socket.java:633)
at java.base/java.net.Socket.connect(Socket.java:583)
at java.base/java.net.Socket.<init>(Socket.java:507)
at java.base/java.net.Socket.<init>(Socket.java:287)
at org.apache.catalina.startup.Catalina.stopServer(Catalina.java:630)
at java.base/jdk.internal.reflect.NativeMethodAccessorImpl.invoke0(Native Method)
at java.base/jdk.internal.reflect.NativeMethodAccessorImpl.invoke(NativeMethodAccessorImpl.java:77)
at java.base/jdk.internal.reflect.DelegatingMethodAccessorImpl.invoke(DelegatingMethodAccessorImpl.java:43)
at java.base/java.lang.reflect.Method.invoke(Method.java:569)
at org.apache.catalina.startup.Bootstrap.stopServer(Bootstrap.java:390)
at org.apache.catalina.startup.Bootstrap.main(Bootstrap.java:479)
```

```
rpavani@C19536 bin % ./startup.sh
Using CATALINA_BASE: /Users/rpavani/Downloads/apache-tomcat-9.0.117
Using CATALINA_HOME: /Users/rpavani/Downloads/apache-tomcat-9.0.117
Using CATALINA_TMPDIR: /Users/rpavani/Downloads/apache-tomcat-9.0.117/temp
Using JRE_HOME: /opt/homebrew/Cellar/openjdk@17/17.0.18/libexec/openjdk.jdk/Contents/Home
Using CLASSPATH:
/Users/rpavani/Downloads/apache-tomcat-9.0.117/bin/bootstrap.jar:/Users/rpavani/Downloads/apache-tomcat-9.0.117/bin/tomcat-juli.jar
Using CATALINA_OPTS:
Tomcat started.
```

```
rpavani@C19536 bin % tail ~/Downloads/apache-tomcat-9.0.117/logs/localhost_access_log*.txt
==> /Users/rpavani/Downloads/apache-tomcat-9.0.117/logs/localhost_access_log.2026-04-17.txt <==
```

```
==> /Users/rpavani/Downloads/apache-tomcat-9.0.117/logs/localhost_access_log.2026-04-20.txt <==
```

```
0:0:0:0:0:0:1 - - [20/Apr/2026:10:11:17 +0530] "GET /asf-logo-wide.svg HTTP/1.1" 200 7089
0:0:0:0:0:0:1 - - [20/Apr/2026:10:11:17 +0530] "GET /bg-nav.png HTTP/1.1" 200 1401
0:0:0:0:0:0:1 - - [20/Apr/2026:10:11:17 +0530] "GET /bg-middle.png HTTP/1.1" 200 1918
0:0:0:0:0:0:1 - - [20/Apr/2026:10:11:17 +0530] "GET /bg-button.png HTTP/1.1" 200 713
0:0:0:0:0:0:1 - - [20/Apr/2026:10:11:17 +0530] "GET /bg-upper.png HTTP/1.1" 200 3103
0:0:0:0:0:0:1 - - [20/Apr/2026:10:12:31 +0530] "GET /jenkins HTTP/1.1" 404 759
0:0:0:0:0:0:1 - - [20/Apr/2026:10:12:31 +0530] "GET /jenkins HTTP/1.1" 404 759
0:0:0:0:0:0:1 - - [20/Apr/2026:10:15:05 +0530] "GET / HTTP/1.1" 200 11232
0:0:0:0:0:0:1 - - [20/Apr/2026:10:16:33 +0530] "GET /jenkins HTTP/1.1" 404 759
0:0:0:0:0:0:1 - - [20/Apr/2026:10:16:34 +0530] "GET /jenkins HTTP/1.1" 404 759
```

```
==> /Users/rpavani/Downloads/apache-tomcat-9.0.117/logs/localhost_access_log.2026-04-22.txt <==
```

```
0:0:0:0:0:0:1 - - [22/Apr/2026:14:11:17 +0530] "GET / HTTP/1.1" 200 11232 "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/147.0.0.0 Safari/537.36"
0:0:0:0:0:0:1 - - [22/Apr/2026:14:11:17 +0530] "GET /tomcat.svg HTTP/1.1" 200 67795 "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/147.0.0.0 Safari/537.36"
0:0:0:0:0:0:1 - - [22/Apr/2026:14:11:17 +0530] "GET /tomcat.css HTTP/1.1" 200 5584 "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/147.0.0.0 Safari/537.36"
0:0:0:0:0:0:1 - - [22/Apr/2026:14:11:17 +0530] "GET /asf-logo-wide.svg HTTP/1.1" 200 7089 "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/147.0.0.0 Safari/537.36"
0:0:0:0:0:0:1 - - [22/Apr/2026:14:11:17 +0530] "GET /bg-nav.png HTTP/1.1" 200 1401 "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/147.0.0.0 Safari/537.36"
0:0:0:0:0:0:1 - - [22/Apr/2026:14:11:17 +0530] "GET /bg-upper.png HTTP/1.1" 200 3103 "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/147.0.0.0 Safari/537.36"
0:0:0:0:0:0:1 - - [22/Apr/2026:14:11:17 +0530] "GET /bg-middle.png HTTP/1.1" 200 1918 "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/147.0.0.0 Safari/537.36"
0:0:0:0:0:0:1 - - [22/Apr/2026:14:11:17 +0530] "GET /bg-button.png HTTP/1.1" 200 713 "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/147.0.0.0 Safari/537.36"
0:0:0:0:0:0:1 - - [22/Apr/2026:14:11:17 +0530] "GET /favicon.ico HTTP/1.1" 200 21630 "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/147.0.0.0 Safari/537.36"
0:0:0:0:0:0:1 - - [22/Apr/2026:14:11:18 +0530] "GET / HTTP/1.1" 200 11232 "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/147.0.0.0 Safari/537.36"
```

```

rpavani@C19536 Scripting Basics: Python-practical task % cd Task-1
rpavani@C19536 Task-1 % python3 log_parser.py
~/Downloads/apache-tomcat-9.0.117/logs/localhost_access_log.2026-04-22.txt
Total unique User Agents: 1

Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/147.0.0.0 Safari/537.36 ->
10

```

3. Develop a python script that:

Creates an rc tag for the latest commit in specific Github repository for main branch

Creates a release tag when executed with a --create-release flag

The rc tag should follow the format vX.Y.Z(e.g: rc/1.2.3). Y increments with each new rc tag created

If run with --create-release flag: Create rc tag and release tag in format X.Y.Z

Output:

```

rpavani@C19536 Task-1 % python3 -m pip install requests
Requirement already satisfied: requests in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages (2.33.1)
Requirement already satisfied: charset_normalizer<4,>=2 in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages (from requests) (3.4.7)
Requirement already satisfied: idna<4,>=2.5 in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages (from requests) (3.11)
Requirement already satisfied: urllib3<3,>=1.26 in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages (from requests) (2.6.3)
Requirement already satisfied: certifi>=2023.5.7 in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages (from requests) (2026.2.25)

[notice] A new release of pip is available: 25.2 -> 26.0.1
[notice] To update, run: python3 -m pip install --upgrade pip
rpavani@C19536 Task-1 % python3 github_tagging.py --repo rpavani-png/Assignments --token
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

Created tag: rc/1.1.0
rpavani@C19536 Task-1 % python3 github_tagging.py \
--repo rpavani-png/Assignments \
--token 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 \
--create-release
Created tag: rc/1.2.0
Created tag: 1.2.0
rpavani@C19536 Task-1 %

```

4. Write a script that gets system information like distro info, memory(total, used, free), CPU info (model, core numbers, speed), current user, system load average, and IP address. Use arguments for specifying resources. (For example, -d for distro -m for memory, -c for CPU, -u for user info, -l for load average, -i for IP address).

Output:

```
rpavani@C19536 Task-1 % python3 -m pip install psutil
Requirement already satisfied: psutil in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages (7.2.2)

[notice] A new release of pip is available: 25.2 -> 26.0.1
[notice] To update, run: python3 -m pip install --upgrade pip
rpavani@C19536 Task-1 % python3 system_info.py -d
Distro: macOS-26.4.1-arm64-arm-64bit-Mach-O
rpavani@C19536 Task-1 % python3 system_info.py -m
Memory -> Total: 25769803776, Used: 7551959040, Free: 5118279680
rpavani@C19536 Task-1 % python3 system_info.py -c
CPU: arm
Cores: 12
Speed: scpufreq(current=4, min=1, max=4)
rpavani@C19536 Task-1 % python3 system_info.py -l
Load Avg: (4.03076171875, 3.06005859375, 2.6982421875)
rpavani@C19536 Task-1 % python3 system_info.py -i
IP: 172.18.153.137
rpavani@C19536 Task-1 % python3 system_info.py -u
rpavani
User: root
rpavani@C19536 Task-1 %
```

Task 2:

Output:

```
rpavani@C19536 Task-2 % python3 survey_script.py survey.json emails.txt

Creating survey: Employee Feedback Survey
Survey created: 422209781
Adding question: How satisfied are you with your workspace?
Adding question: Do you have the tools you need to work effectively?
Adding question: How is communication in your team?
Collector created: 439487067
Survey Link: https://www.surveymonkey.com/r/XCYWDBX

✅ SURVEY CREATED SUCCESSFULLY
Survey ID: 422209781
rpavani@C19536 Task-2 %
```

Can view the created survey here and can answer the questions and submit the form.
Valid for 90 days.

Survey Link: <https://www.surveymonkey.com/r/XCYWDBX>

